

EXP NO : 6

DATE : 09/03/2026

PERFORM SCHEMA DESIGN AND INDEXING

```
>_MONGOSH
> use companyDB
< switched to db companyDB
> db.createCollection("employees")
< { ok: 1 }
> db.employees.insertMany([
  { emp_id: 1, name: "Alice", dept: "HR", salary: 30000, city: "Chennai" },
  { emp_id: 2, name: "Bob", dept: "IT", salary: 40000, city: "Delhi" },
  { emp_id: 3, name: "Charlie", dept: "Finance", salary: 35000, city: "Mumbai" },
  { emp_id: 4, name: "David", dept: "IT", salary: 45000, city: "Chennai" },
  { emp_id: 5, name: "Eve", dept: "HR", salary: 32000, city: "Delhi" }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f6101dc15240977f29ba79'),
    '1': ObjectId('69f6101dc15240977f29ba7a'),
    '2': ObjectId('69f6101dc15240977f29ba7b'),
    '3': ObjectId('69f6101dc15240977f29ba7c'),
    '4': ObjectId('69f6101dc15240977f29ba7d')
  }
}
> db.employees.createIndex({ name: 1 })
< name_1
> db.employees.createIndex({ dept: 1, salary: -1 })
< dept_1_salary_-1
> db.employees.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { name: 1 }, name: 'name_1' },
  { v: 2, key: { dept: 1, salary: -1 }, name: 'dept_1_salary_-1' }
]
> db.employees.find({ name: "Alice" })
< {
  _id: ObjectId('69f605016499d26aa21b37d8'),
  name: 'Alice',
  age: 26,
  dept: 'HR'
}
```

>_MONGOSH

```
  name: 'Alice',
  age: 25,
  dept: 'HR'
}
{
  _id: ObjectId('69f6101dc15240977f29ba79'),
  emp_id: 1,
  name: 'Alice',
  dept: 'HR',
  salary: 30000,
  city: 'Chennai'
}
```

> db.employees.find({ dept: "IT" }).sort({ salary: -1 })

```
< {
  _id: ObjectId('69f6101dc15240977f29ba7c'),
  emp_id: 4,
  name: 'David',
  dept: 'IT',
  salary: 45000,
  city: 'Chennai'
}
```

```
{
  _id: ObjectId('69f6101dc15240977f29ba7a'),
  emp_id: 2,
  name: 'Bob',
  dept: 'IT',
  salary: 40000,
  city: 'Delhi'
}
```

> db.employees.dropIndex({ name: 1 })

```
< { nIndexesWas: 3, ok: 1 }
```

> db.employees.dropIndexes()

```
< {
  nIndexesWas: 2,
  msg: 'non-_id indexes dropped for collection',
  ok: 1
}
```

companyDB>